Identification of Different Types of Buffalo Using YOLO Algorithm Based on Computer Vision

1st Joshua Reska Butu
Departement of Informatics
Hasanuddin University
Makassar, Indonesia
butujr20d@student.unhas.ac.id

2st Ingrid Nurtanio
Departement of Informatics
Hasanuddin University
Makassar, Indonesia
ingrid@unhas.ac.id

3st Ady Wahyudi Paundu
Departement of Informatics
Hasanuddin University
Makassar, Indonesia
adywp@unhas.ac.id

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***Abstract*— The purpose of this research is to create a system that can distinguish and correctly label different types of black and white buffaloes, such as Bonga, Saleko, Lotongboko, and buffaloes that are not black and white or other than their species. This research uses classification methods in pattern recognition, training a model to distinguish the different characteristics of each type of buffalo. This model was then used to classify the buffalo species in the images using the YOLO (You Only Look Once) algorithm. The training involved 40 learning iterations. The dataset consisted of 497 buffalo images, each measuring 640 x 640 pixels, which were divided into 442 images for training and 55 images for validation. By the 40th epoch, the model had achieved an overall mAP50 value of 0.959 and mAP50-95 of 0.825. The research shows that the buffalo species detection and classification system achieved optimal performance, demonstrating the effectiveness of the YOLO algorithm in computer vision, achieving 98% accuracy in the identification and labeling of different buffalo species using image analysis and pattern recognition techniques, while the increasing mAP value indicates that the model is still learning and improving its performance with each additional epoch. This study demonstrates the very high accuracy and potential convergence of the model to optimal performance.**

Keywords—buffalo detection, buffalo pattern recognition, computer vision, YOLO (You Only Look Once)

I. Introduction

The business of buffalo farming presents significant potential as a profitable income source for communities that possess an interest and commitment to buffalo husbandry. In this venture, there are various appealing buffalo breeds to rear, such as Saleko, Bonga, Boko, Balian, Pudu, Todi, Tekken Langi, and others. The uniqueness of these buffaloes can be observed from multiple aspects, such as skin patterns, body color, and horn shapes, all contributing to their market value and appeal. Each buffalo breed exhibits specific characteristics for instance, skin patterns may vary extensively and be strikingly attractive, body colors can be rare and captivating, and the shapes of their horns might be quite unique, garnering high prices in certain markets. These characteristics not only enhance aesthetic value but also can increase the animals' sale value, particularly among traders or for specific purposes such as buffalo contests or traditional ceremonies that involve these animals.

Therefore, the selection of buffaloes is not merely a promising business opportunity but also necessitates knowledge about the diverse types of buffaloes. In various industries, Artificial Intelligence and Computer Vision are often employed. An example of an industry that commonly uses Computer Vision is the livestock security industry. [1] Detection and breed classification of cattle using the YOLO v4 Algorithm." This research addresses a common issue in the recognition of livestock breeds through the application of the 'You Only Look Once' (YOLO) algorithm based on computer vision. With an accuracy rate of 92.85%, this research successfully identified the gender of cattle. However, one identifiable limitation of this study is the scope of detection, which solely considers gender as the primary classification attribute. This results in a restricted number of classes involved in the classification, allowing only two classes (male and female) to be included in the analysis. In other words, this research exclusively focuses on gender classification in cattle, while other attributes such as the identification of more specific buffalo species, morphological characteristics, or specific types of buffalo are not included in the analysis.

II. Literature Review

Research on recognizing different types of buffalo using computer vision, specifically by identifying their body patterns, is not widely conducted. Yet, numerous studies have delved into cattle identification. Some of these have tackled feature extraction methods such as FAST, SIFT, SURF, and ORB, with a particular focus on buffalo. One study titled [2] "Individual identification of Holstein dairy cows based on detecting and matching feature points in body images" focuses on extracting cattle body images to identify Holstein cows. It showcases the effectiveness of using three separate algorithms to detect and match features points for identification. This approach involves video analysis of cows in profile to detect the cattle and identify individual body areas. By comparing unknown images against a database of templates, the system can confirm identities using methods like FAST, SIFT, SURF, and ORB, achieving a high accuracy rate of 96.72%.

The study [3] "Identification of Cattle Breed using the Convolutional Neural Network" examines the categorization of cattle based on size and color, advocating for the use of diverse cattle image datasets for this purpose. It suggests capturing cattle images from multiple angles—front, back, and side—to extract significant features for individual identification. These features are then used to determine the breed of each cattle. The process involves initial image preprocessing, followed by breed categorization, with a special classification for uncommon breeds based on a pre-existing dataset. The Convolutional Neural Network (CNN), a deep learning technique, is employed to detect unique facial and bodily features of the cattle. The research compiled over 150 cattle images, which were preprocessed for uniform dimensions and noise reduction. Then, using the SIFT method, features from various body parts were extracted. These features were classified into 25 different categories with the help of CNN, to predict the breed of the cattle. The system is designed to work with images that display a single animal. Future research directions include breed identification in more complex scenes with multiple animals

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and diverse backgrounds, necessitating additional separation techniques for images containing a mix of cattle breeds.

The study titled [4] "Shape Recognition and Matching by using Contour-SURF and Analysis" explores the use of Contour-SURF and Contour-SIFT for identifying and matching shapes. It proposes a novel system that utilizes Contour-SURF, a method that combines aspects of feature extraction and description, and it benchmarks this system against Contour-SIFT. Results are depicted through graphical analysis across various datasets. Findings reveal that Contour-SURF excels with a 100% accuracy rate in detecting shapes with unique contours, although its performance dips with more generic shapes. Moreover, Contour-SURF is shown to match features more rapidly than Contour-SIFT. The research contributes valuable information about the precision and speed of Contour-SURF in the realm of shape recognition and matching.

This subsequent study employs the YOLOv3 algorithm [5] to classify images of animals, assessing the model's effectiveness with various training and testing images from a selected dataset. The goal is to establish a method for animal detection using YOLOv3, which involves training on a dataset via darknet. The research provides insights into the performance of YOLOv3 in animal identification, noting that the model can make both accurate and inaccurate predictions. Future enhancements include using a specific dataset with a precise image count for each category to improve accuracy. Nevertheless, the current system can sometimes mislabel images or fail to make a prediction. The study observed instances of correct identifications, incorrect ones, and scenarios where the model did not produce any output at all.

Additional investigations into the YOLO algorithm include a study on "Detection and Breed Classification of Cattle Using YOLOv4 Algorithm, [1], " which applies computer vision to categorize cattle breeds by detecting specific body regions unique to each breed. The study's experiments showed that the YOLO algorithm can accurately identify and classify cattle breeds with 92.85% accuracy. Notably, the dataset compiled for this research was categorized based on cattle gender—calves, cows, and bulls—resulting in breed detection being reported alongside gender, which was a significant outcome of the study. Further comparative research examined the performance of YOLOv3 and YOLOv4 [6]. in identifying both individual animals and herds, utilizing unmanned aerial vehicles (UAVs) to capture images of African wildlife. This study, which looked at giraffes, white rhinos, reindeer, and zebras, found that increased altitude could cause loss of image detail, reducing the models' accuracy to 68.86%. In terms of speed, YOLOv4, with 51.70 frames per second (FPS), was faster than YOLOv3, showing a 13.7% improvement. The precision and recall rates of both models were equal at 97% at an altitude of 15m, but YOLOv4 surpassed YOLOv3 by 2% at 20m, and its superiority grew with altitude. Another study [7], engaged YOLOv3 alongside Firebase technology to develop an app for real-time animal tracking, demonstrating YOLO's ability to detect animals in forested environments. The system could accurately identify animals and provide data such as names, timestamps, images, and exact locations. Despite its effectiveness, challenges such as detecting boundaries accurately and the necessity for careful camera placement were noted. This research continued previous work on YOLOv3, addressing additional hurdles like low-light environments and fast-moving animals.

While previous studies have made strides in animal image recognition, there's room for improvement. Study [5] requires enhanced accuracy in animal recognition. Studies [6] and [7], need to refine object detection by factoring in distance and ambient lighting conditions. Unlike these, the study [1] utilized YOLOv4 and focused on identifying cattle breeds, specifically discerning cattle gender as a part of its detection and classification output. The novelty of this research lies in its use of a never-before-applied method, YOLOv8, and its focus on buffalo breeds a subject not previously explored. The application of YOLOv8 aims to advance the detection of various buffalo types, particularly their distinct color patterns, and is anticipated to elevate the precision of buffalo breed recognition significantly.

The difference between this study and previous studies lies in two aspects, namely the object of research and the method used. The object of this research is buffalo, which has not previously been used in research to detect its species. The method used in this research is YOLOv8, which is the latest method and has never been used before to detect buffaloes. This research contributes to the development of a system that is able to accurately distinguish and label various types of buffaloes, such as Bonga, Saleko, Lotongboko, and buffaloes that are not black and white or other types. This research uses classification methods in pattern recognition, training a model to distinguish the different characteristics of each buffalo breed. This model is then used to identify buffalo species in the image using the YOLO (You Only Look Once) algorithm.

III. Proposed Method

This research focuses on creating a system that can recognize different buffalo species through a computer vision technique that incorporates the "You Only Look Once" (YOLO) algorithm. [8] Joseph Redmon, a PhD candidate at the University of Washington, originally developed YOLO, and has since introduced improved iterations like YOLOv2 and YOLOv3. These versions offer substantial enhancements in performance and precision. The YOLO algorithm is designed to become the leading method for object detection within the domains of image processing and computer vision.

The main objective of instance segmentation is to recognize and classify each individual object in an image, such as different types of buffalo, by assigning a label to every pixel linked to that object. This technique enables the differentiation among various buffalo types by generating an image where each pixel is annotated, revealing the identity of the object it represents and specifying the particular buffalo type associated with it.

A. Preprocessing Stage

During the preprocessing phase, multiple steps are implemented. The initial phase consists of data collection, which involves capturing images from various left and right sections of the object. Following this, images are annotated and the YOLOv8 model is trained, utilizing the complimentary GPU resources from Google Colab (see Fig. 1). The assessment of the model's effectiveness relies on deep learning algorithms, using metrics such as model

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Precision, Recall, and Mean Average Precision (mAP) for evaluation.

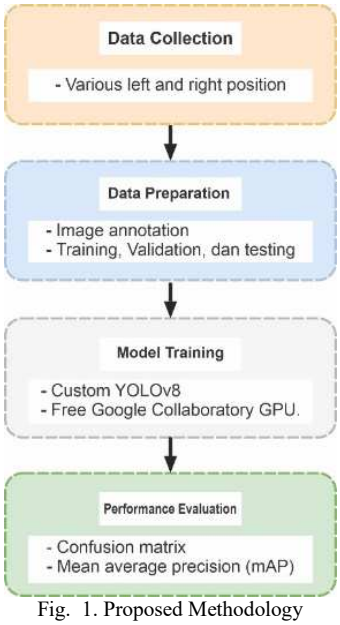

Fig. 1. Proposed Methodology

1. *Data Collection*

At this stage, images of the buffalo were captured using a Samsung Galaxy A52 mobile phone camera, which has a resolution of 4624 X 3468 pixels. The photos were taken from the left and right perspectives, followed by a scaling procedure that shrinks the photos to a dimension of 640 x 640 pixels. The purpose of minimizing the dimensions of the photo is to speed up computational operations.

2. *Data Preparation*

During this stage, subject identification or detection within the images is carried out by applying polygon masks for annotation. Once annotation is finalized, the training dataset is divided into a test set and a validation set. These subsets support the training and assessment stages, respectively. Specifically, 89% of the data is allocated for training, 8% for validation, and 3% for testing, with the division performed via Roboflow software [9]. Initial image processing resulted in a collection of 597 buffalo images, with 442 images intended for training and 40 images for validation purposes. The dataset was organized into four buffalo categories: Saleko, Bonga, Lotongboko, and other unspecified classes. This classification is illustrated in Fig. 2.

3. *The Training*

The training of the YOLOv8 model is carried out by employing the Graphics Processing Unit (GPU), which is available for free via Google's cooperative platforms. This training process involves training iterations known as epochs, with the number set to 40 iterations for the entire dataset.

4. *Performance evaluation*

At the current phase, the effectiveness of the model is assessed. Utilizing advanced deep learning techniques, the evaluation of the model encompasses measures such as model Precision, model Recall, accuracy rate or confusion matrix, and Mean Average Precision (mAP).

Fig. 2. Three Types or Classes of Buffalo : Saleko Buffalo, Bonga Buffalo, and Lotongboko Buffalo.

B. *Training Stage*

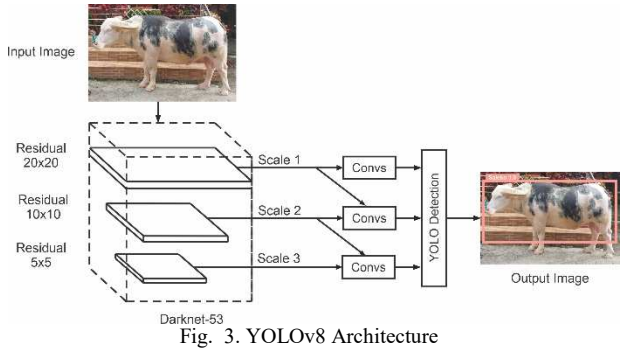

Fig. 3. YOLOv8 Architecture

The YOLO (You Only Look Once) algorithm was originally proposed by computer vision researcher Joseph Redmon [8]. It begins by processing buffalo images at 640x640 pixels. At its core, a deep learning neural network extracts features using convolutional layers within the Darknet framework [10]. The process involves segmenting the image into a grid, with each segment tasked with object prediction. YOLO employs a set number of anchor boxes per grid segment to predict the position and dimensions of objects. The detection is finalized by forecasting various essential elements for each anchor box within the grid cells (see Fig. 3).

C. *Performance Evaluation*

The picture illustrates a pair of assessment tools used for gauging the effectiveness of predictive models. A confusion matrix provides a visual layout that categorizes prediction results into correct and incorrect classifications. On the other hand, Mean Average Precision (mAP) quantitatively measures the accuracy of identifying objects, factoring in different categories and decision boundaries, a standard in the realm of computer vision. These tools are essential for interpreting and refining the object recognition proficiency of a model.

1. *Model Evaluation*

Mean Average Precision (mAP) is an evaluation metric that measures the quality of object detection across multiple classes. The average precision for every category is determined by

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evaluating the precision-recall curve's area while considering the Intersection over Union (IoU) metric. Once the average precision is computed for each class, the mean of these average precision values is taken.

$$mAP = \frac{1}{N} \sum APi \quad (1)$$

In the given equation, N (1) denotes the sum of all classes, while AP refers to the average precision. Thus, mAP (1) assesses the ability of a system to accurately identify objects across a range of classes [6].

2. *Precision and Recall*

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

In this equation, Precision is calculated as the fraction of True Positives (*TP*) over the combined number of True Positives and False Positives (FP), expressed as Precision = *TP* / (*TP* + *FP*) (2). This measure indicates the proportion of the model's positive predictions that are accurate. On the other hand, Recall, also referred to as Sensitivity or the True Positive Rate, quantifies a model's ability to identify all real positive cases. It is computed by dividing the number of True Positives (*TP*) by the total of True Positives and False Negatives (*FN*), with the formula Recall = *TP* / (*TP* + *FN*) (3). This metric assesses the model's proficiency in detecting or "recalling" all actual positives.

3. *Confusion Matrix*

A confusion matrix serves as a metric for assessing the accuracy of a classification algorithm. For binary classification, it consists of four elements: True Positives (*TP*), True Negatives (*TN*), False Positives (*FP*), and False Negatives (*FN*). These components reflect the count of accurate and inaccurate forecasts the model makes across the two classes. From this matrix, we can calculate important metrics such as accuracy, precision, and recall. For multi-class classification with many classes, the confusion matrix is expanded into an n x n table, with n representing the number of classes. The diagonal values represent correct predictions, while the off-diagonal values indicate classification errors among classes. The confusion matrix is very useful because it provides information not only on the model's overall success but also on the types of mistakes made, allowing for further refinement of the model. This is particularly important in applications where certain types of errors may have more severe consequences than others [11].

IV. Results and Discussion

During the training phase of the model, which utilized a dataset containing 442 buffalo images, the training was conducted over 40 training cycles or epochs. As a result, the model achieved a mean Average Precision (mAP) score at a detection threshold of 50% (mAP50) of 0.964, while at a detection threshold ranging from 50% to 95% (mAP50-95), it achieved a score of 0.862. The determination of the optimal number of epochs at 40 and the setting of the step interval per epoch was based on the observation of progressive improvement of the mAP values during the training process. TABLE I presents a comparison of the mAP50 with the mAP50-95 at each epoch, along with detailed steps of the training evolution. The increasing mAP values indicate that the model continued to learn and enhance its performance with each additional epoch. At the 40th epoch, the model had reached an mAP50 of 0.959 and an mAP50-95 of 0.825, indicating a very high level of accuracy and the model's potential convergence to optimal performance.

TABLE I. Comparison of mAP50 with mAP50-95 at each Epoch and Epoch Steps.

No	Epoch	mAP50	mAP50-95
1	10	0.566	0.453
2	20	0.629	0.533
3	30	0.813	0.691
4	40	0.959	0.825

Furthermore, the results of precision and recall demonstrate the performance of the model trained over 100 epochs on various buffalo classes such as bonga, lotongboko, saleko, and others. TABLE II displays the mAP50, and TABLE III presents the mAP50-95 throughout the training process per class at various epochs.

TABLE II. Result of mAP50 on bonga, lotongboko, saleko and other classes

Epoch	Class			
	bonga	*lotongboko*	*saleko*	*othertypes*
10	0.726	0.199	0.701	0.638
20	0.975	0.142	0.895	0.503
30	0.948	0.497	0.838	0.995
40	0.939	0.995	0.906	0.995

TABLE III. Result of mAP50-95 on bonga, lotongboko, saleko and other classes

Epoch	Class			
	bonga	*lotongboko*	*saleko*	*othertypes*
10	0.453	0.139	0.521	0.56
20	0.858	0.569	0.764	0.452
30	0.893	0.348	0.771	0.751
40	0.939	0.597	0.861	0.902

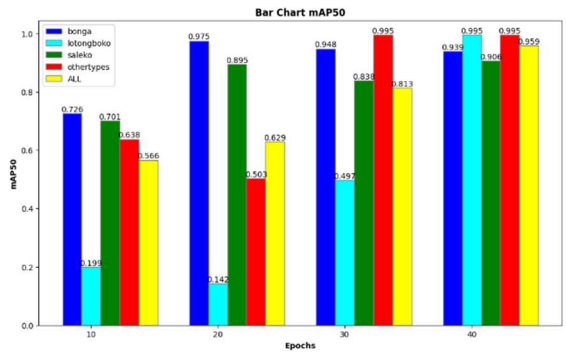

Fig. 4. mAP50 bar chart

It can be observed that the training model from epoch 10 to 40, the bonga buffalo class achieved an mAP50 of 0.939 and mAP50-95 of 0.939, the lotongboko buffalo class

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achieved an mAP50 of 0.995 and mAP50-95 of 0.597, the saleko buffalo class had an mAP50 of 0.906 and mAP50-95 of 0.861, and the other classes achieved an mAP50 of 0.861 (see Fig. 4 and Fig. 5). 995 and mAP50-95 of 0.597, the saleko buffalo class had mAP50 of 0.906 and mAP50-95 of 0.861, and the other classes had mAP50 of 0.995 and mAP50-95 of 0.902.

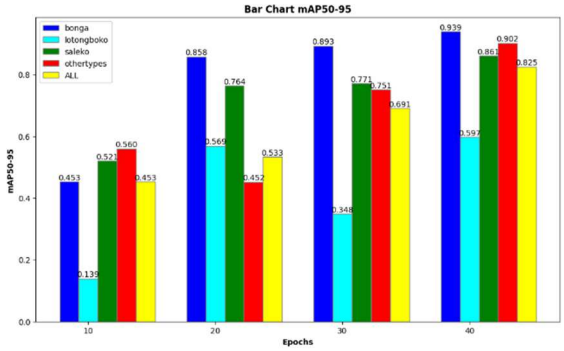

Fig. 5. mAP50-59 Bar Chart.

The 4x4 confusion matrix results (Fig. 6) illustrate the classification model's performance in differentiating between four distinct classes: bonga, lotongboko, saleko, and other classes. From the matrix, it is evident that the model is highly capable of identifying bonga with a perfect score of 1.00, lotongboko with 1.00, other classes with 1.00, and for the saleko class, the model was successful in identifying its category with an accuracy of 0.81. Below is the image of the confusion matrix, where the value of 0.81 in the cell (saleko, saleko) of the confusion matrix indicates that 81% of the actual "saleko" data was correctly predicted by the model as "saleko". The model accurately identified 81% of them, while 19% of the data was misclassified into other categories such as "bonga", and "lotongboko".

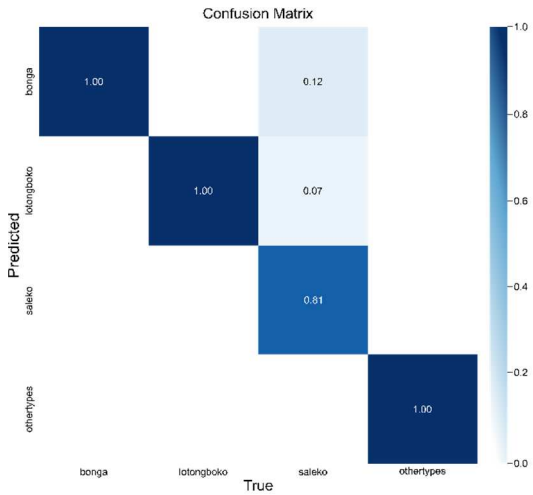

Fig. 6. Confusion Matrix.

The following represents the training data consisting of 442 buffalo images with automatic orientation variations as well as normal and tilted positions (Fig. 7). The results from this set of training data can be examined in the accompanying illustration.

Fig. 7. Training data Result.

The following presents the results of buffalo type recognition capable of categorizing buffalo into several classes, namely: bonga buffalo, lotongboko buffalo, saleko buffalo, as well as other classes that do not include the three aforementioned types (see Fig. 8). The accompanying illustration depicts the data from the recognition of these buffalo species.

Fig. 8. Recognition Results of Bonga Buffalo, Lotongboko Buffalo, and Saleko Buffalo.

V. Conclusion

The results indicate that the buffalo detection and identification system achieved optimal performance with an accuracy range of 98%. This is evidenced by the overall Mean Average Precision (mAP) value on the high test data which is 0.959. Given this context, research on high-accuracy identification of buffalo types opens significant opportunities for the development of practical applications. A primary recommendation is the implementation of computer vision technology for buffalo price estimation. This application can automatically estimate prices based on

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the identified characteristics of the buffalo, presenting an efficient and accurate solution for breeders and traders. Furthermore, the development of a computer vision-based application specifically designed to assist in the identification and evaluation of buffalo types would be highly beneficial. This application would not only improve the accuracy of evaluations but also provide transparent pricing information. To ensure the success and effectiveness of this technology, it is crucial to involve various stakeholders such as breeders, traders, and industry experts in the development process. Thus, the results of this research can be applied on a broader scale and significantly contribute to the development of technology for buffalo price estimation, supporting the overall advancement of the buffalo farming industry.

REFERENCES

[1] A. Yılmaz, G. N. Uzun, M. Z. G¨urb¨uz, and O. Kıvrak, "Detection and breed classification of cattle using yolo v4 algorithm," in 2021 International Conference on INnovations in Intelligent SysTems and Applications (INISTA). IEEE, 2021, pp. 1–4.

[2] K. Zhao, X. Jin, J. Ji, J. Wang, H. Ma, and X. Zhu, "Individual identification of holstein dairy cows based on detecting and matching feature points in body images," Biosystems Engineering, vol. 181, pp.128–139, 2019.

[3] S. Manoj, S. Rakshith, and V. Kanchana, "Identification of cattle breed using the convolutional neural network," in 2021 3rd International Con-ference on Signal Processing and Communication (ICPSC). IEEE, 2021, pp. 503–507.

[4] B. Keerthishree and M. Amed, "Shape recognition and matching by using contour-surf and analysis," in 2018 International Conference on Inventive Research in Computing Applications (ICIRCA). IEEE, 2018, pp. 550–554.

[5] B. K. Reddy, S. Bano, G. G. Reddy, R. Kommineni, and P. Y. Reddy, "Convolutional network based animal recognition using yolo and dark-net," in 2021 6th international conference on inventive computation technologies (ICICT). IEEE, 2021, pp. 1198–1203.

[6] T. Petso, R. S. Jamisola, D. Mpoeleng, and W. Mmereki, "Individual animal and herd identification using custom yolo v3 and v4 with images taken from a uav camera at different altitudes," in 2021 IEEE 6th Inter-national Conference on Signal and Image Processing (ICSIP). IEEE, 2021, pp. 33–39.

[7] S. Prabhu, M. Sreenath, V. Malavika, H. Om, and S. Swetha, "Detectionand recognition of animals using yolo algorithm," in 2023 International Conference on Distributed Computing and Electrical Circuits and Elec-tronics (ICDCECE). IEEE, 2023, pp. 1–6.

[8] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, pp. 779–788.

[9] F. Ciaglia, F. S. Zuppichini, P. Guerrie, M. McQuade, and J. Solawetz, "Roboflow 100: A rich, multi-domain object detection benchmark," arXiv preprint arXiv:2211.13523, 2022.

[10] T. Diwan, G. Anirudh, and J. V. Tembhurne, "Object detection using yolo: Challenges, architectural successors, datasets and applications,"multimedia Tools and Applications, vol. 82, no. 6, pp. 9243–9275, 2023.

[11] I. Markoulidakis, G. Kopsiaftis, I. Rallis, and I. Georgoulas, "Multi-class confusion matrix reduction method and its application on net promoter score classification problem," in The 14th pervasive technologies related to assistive environments conference, 2021, pp. 412–419.

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